

Properties of Quadratic Functions

For

$$f(x) = ax^2 + bx + c, \quad a \neq 0,$$

the graph is a parabola with a vertical axis of symmetry.

Feature	Formula or condition	Meaning
Opening	$a > 0$ or $a < 0$	up for $a > 0$; down for $a < 0$
Axis of symmetry	$x = -\frac{b}{2a}$	vertical line through the vertex
Vertex	$\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$	minimum if $a > 0$; maximum if $a < 0$
y-intercept	$(0, c)$	value of the function at $x = 0$

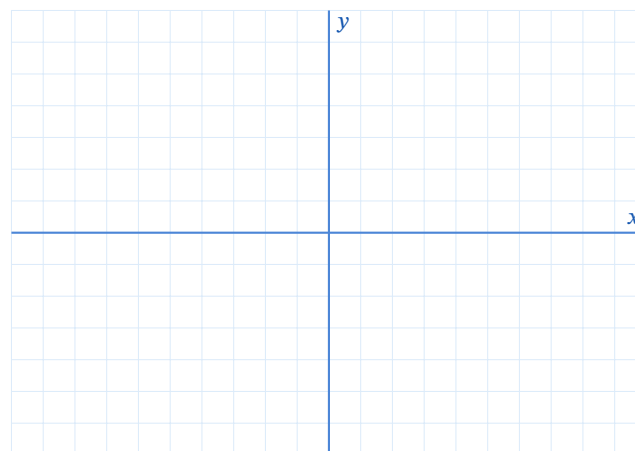
The Squaring Function

The elementary quadratic function is

$$f(x) = x^2.$$

- vertex: $(0, 0)$;
- axis of symmetry: $x = 0$;
- domain: $(-\infty, \infty)$;
- range: $[0, \infty)$;
- minimum value: 0;
- continuous on its domain.

Plot at least five points and sketch the parabola.



Find the Vertex from Standard Form

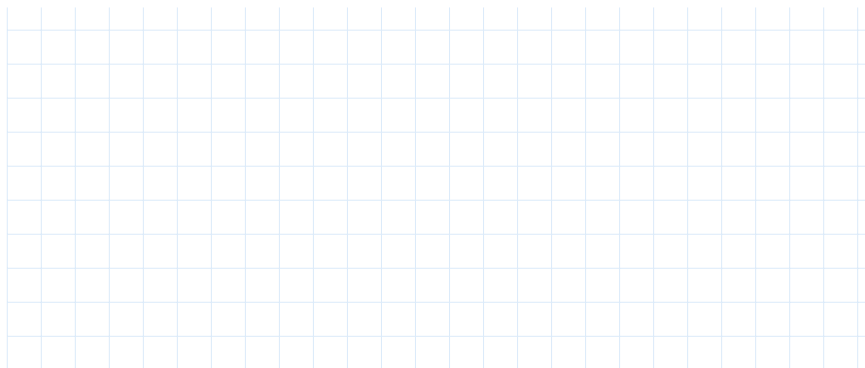
For $f(x) = ax^2 + bx + c$, first find

$$h = -\frac{b}{2a}, \quad k = f(h).$$

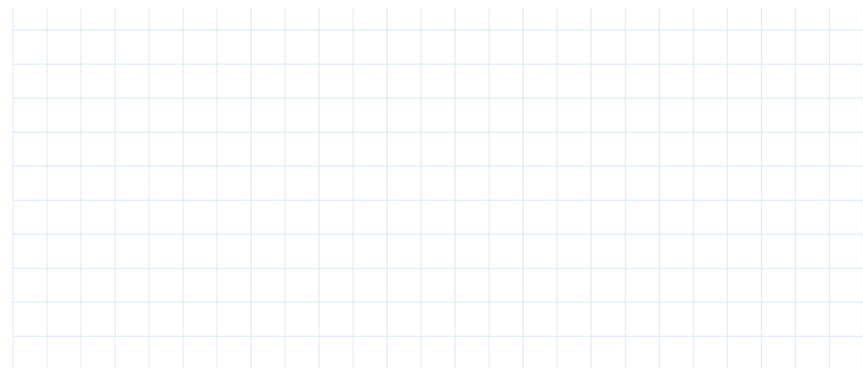
Then the vertex is (h, k) and the axis of symmetry is $x = h$.

Find the vertex and axis of symmetry of each function.

$$f(x) = x^2 - 4x + 5$$



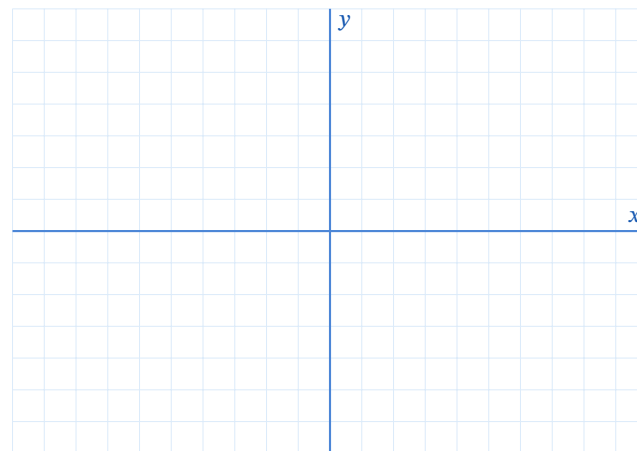
$$g(x) = x^2 - x - 6$$



Graph from Standard Form

Graph

$$f(x) = 2x^2 + 4x + 5.$$



State each feature.

- opening direction;
- vertex;
- axis of symmetry;
- y-intercept;
- domain and range;
- maximum or minimum value.

Vertex Form and Shifts

In vertex form,

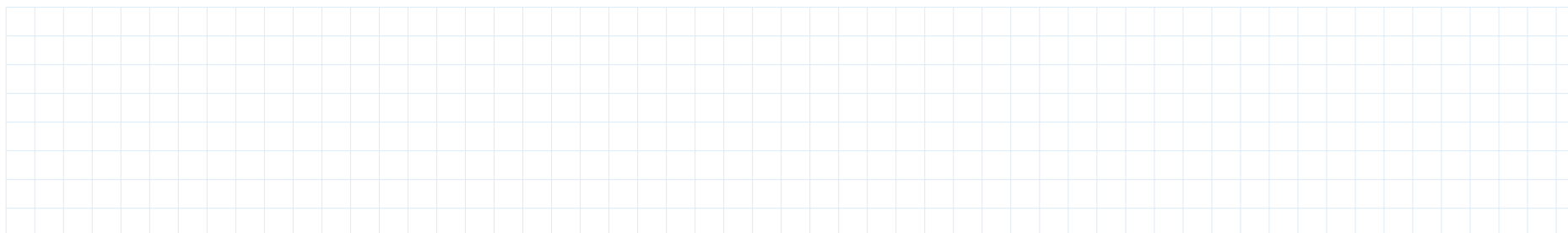
$$f(x) = a(x - h)^2 + k,$$

the vertex is (h, k) and the axis of symmetry is $x = h$.

Starting with $y = x^2$:

- h gives the horizontal shift;
- k gives the vertical shift;
- the sign of a gives the opening direction;
- $|a|$ controls vertical stretch or compression.

Describe and sketch $g(x) = -2(x + 1)^2 + 3$.



Build a Quadratic Function

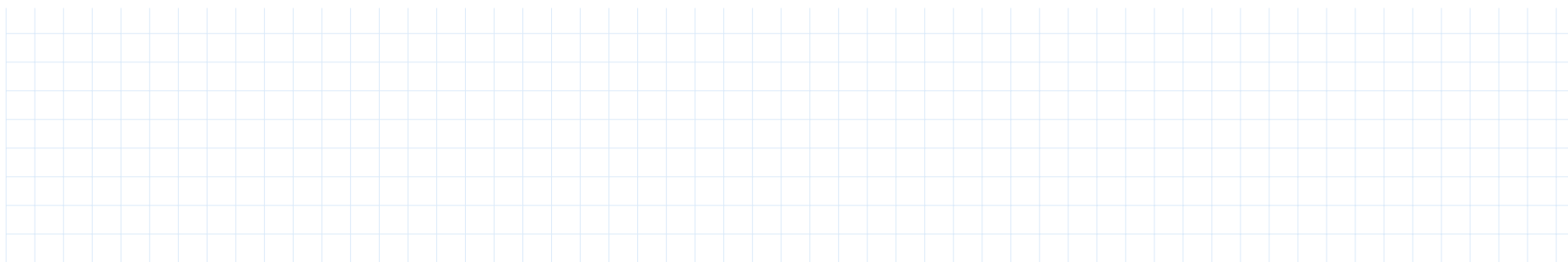
If the vertex (h, k) and another point (x_1, y_1) are known, begin with

$$f(x) = a(x - h)^2 + k.$$

Substitute the known point to solve for a .

Example. Find the quadratic function with vertex $(1, -5)$ and y -intercept -3 .

1. Substitute $(h, k) = (1, -5)$ into vertex form.
2. Use the point $(0, -3)$ to find a .
3. Write the final function and verify the point.



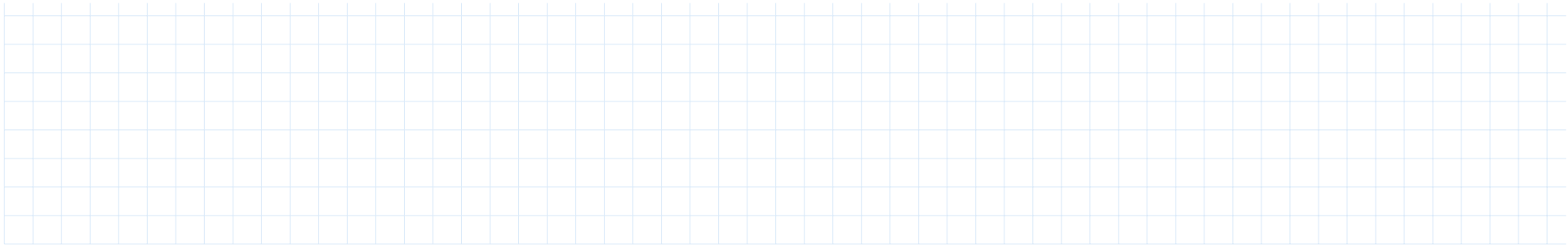
Maximum and Minimum Values

The vertex gives the extreme value of a quadratic function.

Condition	Extreme value	Where it occurs
$a > 0$	minimum value k	at $x = h$
$a < 0$	maximum value k	at $x = h$

For $P(t) = -5t^2 + 40t + 12$:

- (a) Does the graph have a maximum or minimum?
- (b) At what value of t does it occur?
- (c) What is the extreme value, and what units should it have?



Optimization: Enclose the Most Area

David has 400 yards of fencing for a rectangular field. Let w be the width.

- (a) Express the length in terms of w .
- (b) Write the area $A(w)$ as a quadratic function.
- (c) Find the value of w that gives the largest area.
- (d) Find the corresponding length and maximum area.
- (e) State a reasonable domain for the model.

